

Concise Biosynthesis of Antifungal Papulacandins

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ABSTRACT: We report the complete biosynthesis of antifungal papulacandins through heterologous expression and enzymatic assays. The papulacandin biosynthesis follows a convergent five-component strategy, involving a linear polyketide chain that is synthesized and installed to the aryl-glucoside by a polyketide synthase fused with a C-terminal acyltransferase domain. The formation of the challenging tricyclic benzannulated spiroketal core is initiated by the C-glycosylation of 5-(hydroxymethyl)-resorcinol, followed by spirocyclization catalyzed by a Fe(II)/ α -ketoglutarate-dependent oxygenase PpcE.

The papulacandins are a family of naturally occurring antifungal agents initially isolated from the deuteromycete *Papularia sphaerosperma* in 1976.¹ Papulacandins inhibit β -(1,3)-D-glucan synthase (GS), an essential enzyme responsible for the biosynthesis of dominant glucan in fungal cell walls.² The IC₅₀ value of papulacandin B toward GS is as low as 0.02 μ g/mL, which is 10-fold more potent than the first-in-class antifungal drug caspofungin.³ Papulacandins are structurally characterized by a 1,7-dioxaspiro-[5.4]decane skeleton with an aryl- β -D-C-glycopyranoside derived from 5-(hydroxymethyl)resorcinol, and acylated at positions 3 and 6' with residues of long and medium chain fatty acids, respectively (Figure 1). Other papulacandin congeners vary with respect to the diversity of fatty acids and the forms of ring-opened spiroketal.⁴ Because of the potent antifungal activity and the unique structure, papulacandins have attracted significant interest as synthetic targets.⁵ The benzannulated spiroketal unit, a key determinant of the fungicidal activity of papulacandins, poses one of the most provocative challenges in chemical synthesis. The main strategies to construct the spiroketal moiety include the hetero Diels–Alder reaction,⁶ the palladium (0)-catalyzed coupling,⁷ and the *m*-CPBA mediated oxidative spiroacetalization.^{8a} So far, only the total synthesis of papulacandin D (2), the simplest member of the family, was reported in 31 steps and less than 9.2% overall yield.⁸ These findings prompted us to investigate the biosynthesis of papulacandin and to understand how nature performs these challenging transformations.

Toward this goal, we genome-sequenced the papulacandin (L-687,781, 1) producing fungus *Codinaeaella minuta* ATCC 20960 (Previously *dictyochoeta simple*).^{9,10} In line with the convergent total synthesis of 2, we hypothesize that the biosynthesis of 1 proceeds by a convergent five-component strategy, involving an aromatic ring, a glucose, a galactose, a medium-chain fatty acid, and a long-chain fatty acid (Figure 1). Therefore, the biosynthetic gene cluster should encompass genes encoding a nonreducing polyketide synthase (NRPKS), glycosyltransferase, and highly reducing polyketide synthase (HRPKS). Using the NRPKS orsellinic acid synthase ArmB¹¹

as a query to search the whole genome, hits were refined by searching for nearby glycosyltransferase and HRPKS encoding genes. We identified a gene cluster (named *ppc*) containing 10 biosynthetic genes (Figure 2A, Table S3). These genes encode a NRPKS (PpcA), a decarboxylase (PpcB), two P450 monooxygenases (PpcC and PpcI), two glycosyltransferases (PpcD and PpcG), an Fe (II)/ α -ketoglutarate dependent oxygenase (α KGD) (PpcE), an HRPKS with a terminal carnitine acyl-transferase-like domain (PpcF), an acyltransferase (PpcH), and a major facilitator superfamily (MFS) (PpcJ). An NCBI blast identified more than 80 *ppc* homologous clusters from diverse plant pathogenic fungi, and each minimally encodes homologues to five of the genes (*ppcA-D*, and *ppcF*) (Figure S2).

To assess the functions and products of the gene cluster, we heterologously expressed the *ppc* genes in *Aspergillus nidulans* A1145 followed by metabolite isolation and structural characterization.¹² PpcA shows 38% sequence identity to ArmB, which produces the orsellinic acid (3) in melleolides biosynthesis.¹¹ Expression of PpcA alone yielded 3 (Figure S4 for heterologous expression of *ppcA-C*, Figures S20–21). PpcB and PpcC shows 34% and 38% sequence identity to YanB and YanC, respectively.¹³ YanB catalyzes the decarboxylation of methylsalicylic acid to yield cresol, which is subsequently hydroxylated by YanC to produce toluquinol during yanuthone biosynthesis. Coexpression of PpcA with PpcB produces 3,5-dihydroxytoluene (4) (Figure S4, Table S7, Figures S22–S23). Coexpressed PpcABC produces 3,5-dihydroxybenzyl alcohol (5) (Figure S4, Table S8, and Figures S24–S25).

With 5 in hand, PpcD and PpcG are plausible candidates for aryl C-glucosyltransferase. We then expressed and purified recombinant versions of PpcD and PpcG in *E. coli* (Figure S6).

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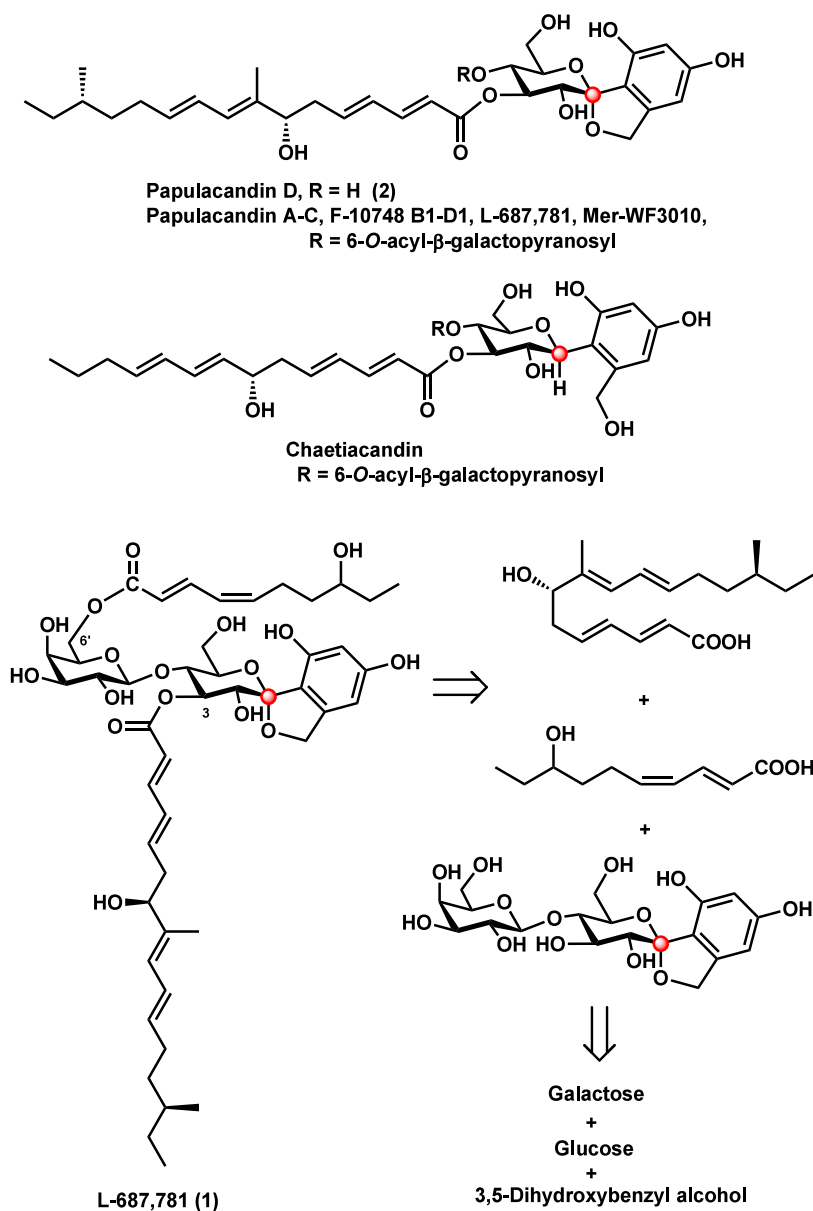


Figure 1. Structures of papulacandins and retrobiosynthetic analysis of L-687,781.

When 5 μM of PpcD was added to 1 mM of **5** and 2 mM UDP-glucose in sodium phosphate buffer, **6** was formed, the $k_{\text{cat}}/K_{\text{M}}$ value is $246.7 \text{ mM}^{-1} \text{ s}^{-1}$ (Figure 2D, Figure S7 for kinetic analysis of PpcD). NMR and HRMS characterization revealed the structure of **6** (Table S9, Figures S26–S31), suggesting PpcD is an aryl C-glucosyltransferase. Unlike other reported phenol C-glucosyltransferases, which typically glycosylate the carbon bearing two vicinal hydroxyl groups,¹⁴ PpcD regiospecifically glycosylates the C2 which is substituted with a hydroxyl group and an alkyl group at the vicinal positions (Figure 2B). The exclusive β -configuration of the C–C bond at the anomeric center of **6** indicates that PpcD catalyzes regio- and stereoselective nucleophilic addition via $\text{S}_{\text{N}}2$ -like process.^{15,16} The remaining glycosyltransferase, PpcG, is, therefore, likely an O-galactosyltransferase. **6** is a versatile intermediate that undergoes three independent reactions at three individual positions to yield **1** ultimately. These reactions include esterification at OH-3 by an unsaturated fatty acid, glycosylation at OH-4, and spirocyclization between the

anomeric carbon and aryl alcohol. To determine the subsequent reactions, we feed commercially available **5** to recombinant *A. nidulans* expressing PpcD together with the remaining enzymes (PpcE, PpcF, PpcG and PpcH) individually. Unexpectedly, distinct traces emerged in both PpcD/PpcE and PpcD/PpcF recombinant strains, respectively (Figure 2C, Figure S5 for heterologous expression of *ppcD-F*). The new compound **7** in the PpcD/PpcE recombinant strain has a molecular weight (MW) of 300 Da, which is 2D smaller than that of **6**, corresponding to the loss of two hydrogen atoms. NMR and HRMS confirmed **7** to be deacylated papulacandin D (Table S10, Figures S32–S37). The configuration of the benzannulated spiroketal was further confirmed by comparison of the chemical shift of C-1 with known papulacandins. PpcE is thus a spirocyclase. To uncover the enzymatic mechanism of PpcE, we expressed recombinant PpcE with N-terminal His-tag in *E. coli*. When 10 μM of PpcE was added to 1 mM of **6** in Tris-HCl reaction buffer at pH 7.5, in the presence of 2 mM αKG , 10 μM $(\text{NH}_4)_2\text{FeSO}_4$, and 0.1 mM ascorbic acid, **7** was

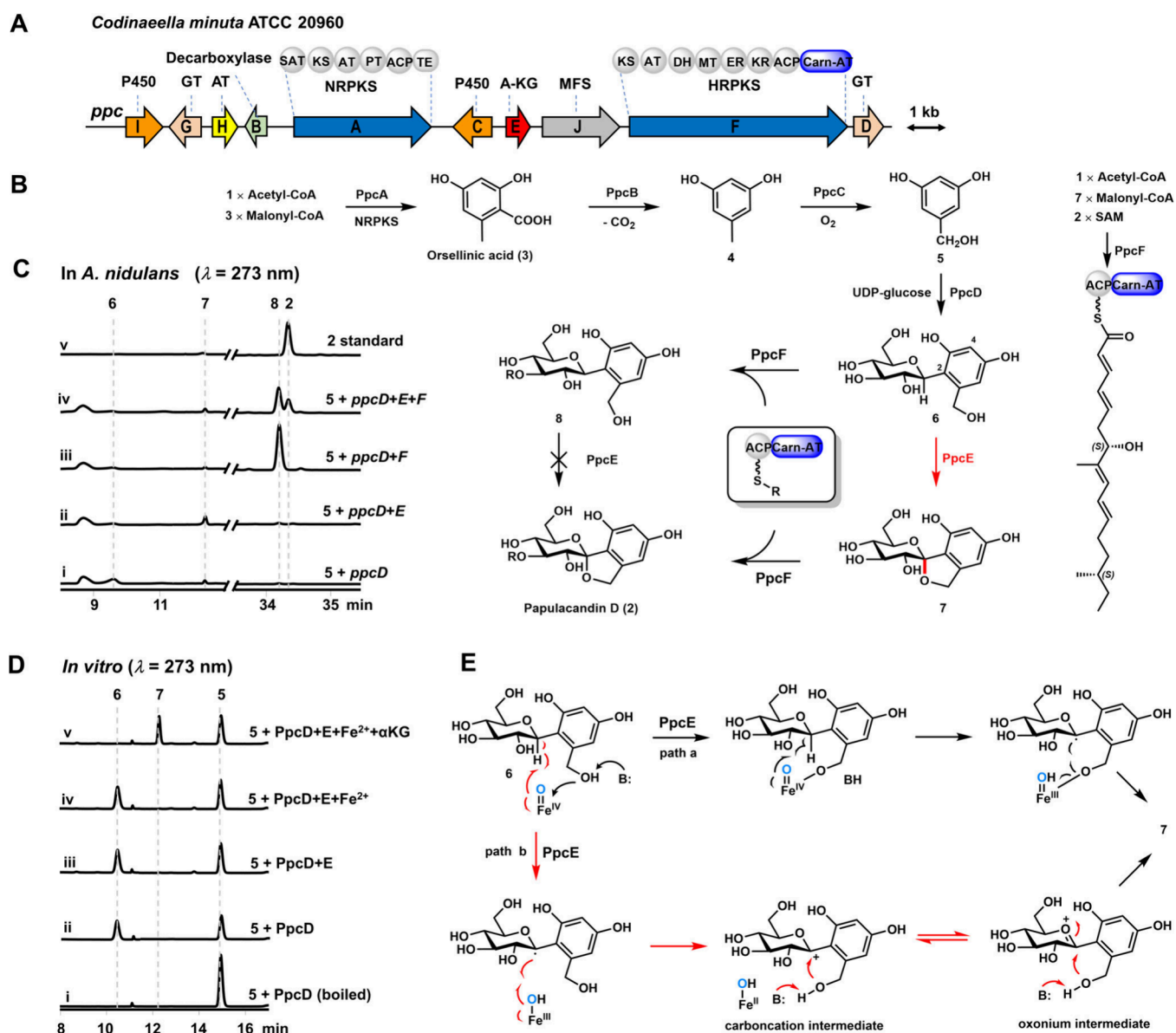


Figure 2. Characterization of L-687,781 (1) biosynthesis. (A) The *ppc* cluster encodes PpcA–PpcJ; (B) The proposed biosynthetic pathway to 2; (C) HPLC analysis of products profiles of heterologous expression of different combinations of *ppc* genes in *A. nidulans* A1145; (D) *In vitro* characterization of PpcD and PpcE; (E) Proposed mechanisms for the spirocyclization reaction catalyzed by PpcE.

formed. The k_{cat}/K_M value is $29.7 \text{ mM}^{-1} \text{ s}^{-1}$ (Figure 2C, Figure S8 for kinetic analysis of PpcE). The absence of αKG or Fe (II) nearly abolished the production of 7, indicating PpcE is both iron and αKG dependent (Figure 2D, Figure S9 for αKG dependence experiment).

No function characterized homologues hits of PpcE were identified, further phylogenetic analysis suggested that PpcE was associated with halogenase BesD and oxidase HygX (17% sequence identity) (Figure S10).^{17,18} To investigate the molecular mechanism of the promiscuous PpcE, we created a PpcE model using alphafold2.¹⁹ The model shows that PpcE exhibits the double stranded β -helix fold (see Figure S11A for PpcE model structure). Adding iron and αKG into the active cavity by Autodock shows that iron and αKG are bound well, as found in most Fe/ αKG hydroxylases by a conserved 2-His-1-Asp (His181, His279, and Asp183) facial triad (Figure S11Ba).²⁰ Docking 6 into the model revealed that 6 is located on the opposite side of the facial triad in an almost identical

fashion. The distance between the iron and the initial reaction site C1 hydrogen of 6 is 3.8 Å, which is close enough for hydrogen atom abstraction by the C–H bond activation (Figure S11Bb). Further mutagenetic analysis identified residues involved in the spirocyclization (Figure S11Bd), with the W242F mutation nearly abolishing bioactivity compared with the wild type. The 4.1 Å distance between the nitrogen of W242 and the hydrogen of phenolic hydroxyl indicates W242 probably acts as a base to abstract the proton of phenolic hydroxyl (Figure S11Bc). Two possible mechanisms are proposed for PpcE (Figure 2E), inspired by the mechanisms of oxacyclase LoLO and desaturase PtID.^{21,22} we propose that the spirocyclization reaction is initiated by H atom at C1 abstraction by the Fe (IV)–oxo complex to form an Fe (III)–OH species and a substrate radical. In pathway (a), the substrate radical was captured directly by the benzylalcohol facilitated by a base residue, probably W242, to give 7; in pathway (b), the electron is transferred from the

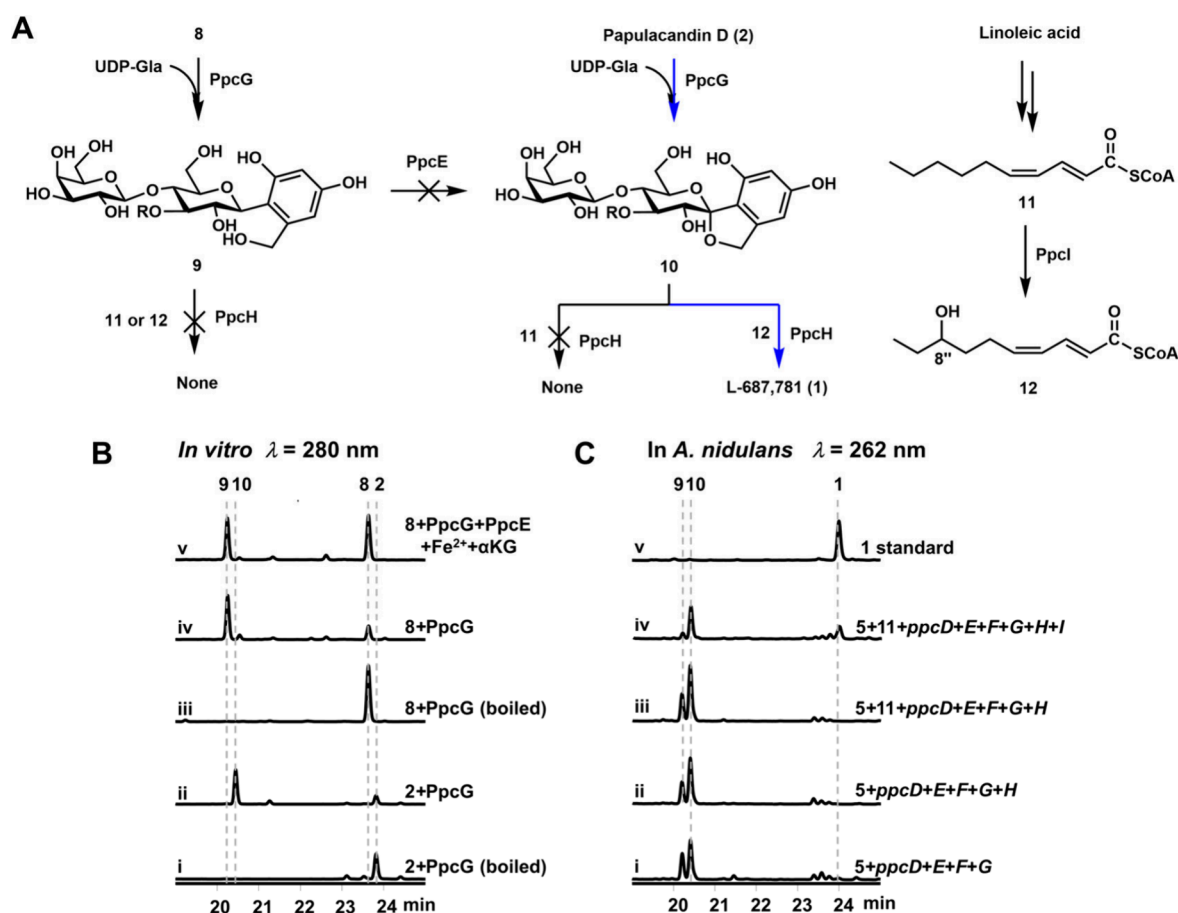


Figure 3. (A) The proposed biosynthetic pathway from 2 to 1; (B) In vitro characterization of PpcG. (C) HPLC analysis of product profiles of heterologous expression of different combinations of *ppcD-I* in *A. nidulans* A1145.

substrate radical to the hydroxoiron(III) cofactor, generating a carbocation at the anomeric carbon, which is followed by intramolecular nucleophilic addition by the benzylic hydroxyl to give 7 (Figure 2B). A possible OH rebound mechanism is also proposed (Figure S12). α KGDs usually catalyze remarkably diverse C–H activation reactions including hydroxylation, halogenation, epoxidation, desaturation, ring expansion/contraction et al.^{20,23} However, α KGD is rarely involved in spirocyclization,¹⁸ our result expanded the repertoire of reactions of α KGDs. Further research is needed to confirm the catalytic mechanism of PpcE.

The new compound 8 from the PpcD/PpcF recombinant strain was confirmed to be a 3-OH acylated acyl glycoside (Table S11, Figures S34–S39). In contrast to 6, 8 cannot be spirialized by PpcE to afford 2 (Figure 3B). Whereas, feeding 5 to a PpcD-F expressing strain yields 2 (Figure 2D), indicating that PpcE and PpcF work sequentially to produce 2. The atypical HRPKS PpcF has a C-terminal domain with significant homology to carnitine *O*-acyltransferase (cAT). To date, only two cAT fused HRPKSs (Tv6-931 and LtbA) have been characterized. The cAT domain of Tv6-931 esterifies the polyketide product by polyalcohol nucleophiles to facilitate complete gem-dimethylation.²⁴ The cAT domain of LtbA utilizes glucinol as a substrate to release the polyketide chain to yield a carbocyclic sugar-containing glycoside.²⁵ PpcF is the first example of a polyketide synthase that esterifies its polyketide chain to an aryl C-glycoside directly.

We next aimed to complete the biosynthetic pathways from 2 to the other papulacandins. With PF-10748 A1 in hand,²⁶ it is evident that 2 undergoes further galactosylation. The remaining glycosyltransferase PpcG is thus the candidate. When 5 μ M PpcG was added to 1 mM of 2 and 2 mM of UDP-galactose in sodium phosphate buffer, near complete conversion of 2 to 10 was observed (Figure 3B) (Table S13, Figures S46–S47). In addition to 2, PpcG can also galactosylate 8 to yield 9 (Figure 3B) (Table S12, Figures S40–S45). Galactosyltransferases are essential to generate galactose containing glycans, glycolipids, and glycoproteins in mammals. However, it is rarely involved in fungal natural product biosynthesis. Our result expanded galactosyltransferases in the repertoire of enzymes in fungal natural product biosynthesis.

In the papulacandin family of antifungals, a majority of compounds including papulacandin A-C, L-687,781, Mer-WF-3010²⁷ and F-10748 B1-D1, have their OH-6 of galactose acylated, which are essential for the penetration of papulacandins into fungal pathogens. These side chains range between butyryl and decanoyl, with deca-2,4-dienoyl and its derivatives being the most abundant (see Figure S1 for a summary of papulacandin family compounds). Deca-2,4-dienoyl containing natural products were rarely found in nature.²⁸ To the best of our knowledge, unsaturated fatty acids including linoleic acid can be degraded by β -oxidation to yield medium chain fatty acid.²⁹ The lack of additional HRPKS in *ppc* implied that the deca-2,4-dienoyl in 1 is probably derived

from linoleate. The acyltransferase PpcH is thus responsible for final acylation. The remaining P450 PpcI, which is exclusive to *ppc*, is likely responsible for the hydroxylation of C-8". After many attempts, we observed **1**, which is produced only in *A. nidulans* expressing PpcD-I in the presence of **5** and external (2E,4Z)-deca-2,4-dienoic acid. We thus propose that PpcI catalyzes C8" hydroxylation of **11** to yield **12**, and membrane protein PpcH then catalyzes esterification between **12** and OH-6 of galactose of **10** to yield **1** (Figure 3C). Scale up fermentation followed by NMR and HRMS characterization confirmed the structure of **1** (Table S4, Figures S15–S16).

In conclusion, we reconstituted the biosynthesis of **1** and established the biosynthetic relationships between the papulacandin family of compounds. Several enzymes that catalyze the synthetic challenging steps are uncovered, and can serve as useful biocatalytic tools to achieve more robust synthesis of papulacandin family of antifungals.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/jacs.4c13101>.

Experimental details, spectroscopic and computational data. Tables S4–S13 are spectroscopic data for compounds **1**–**10**. Figures S15–S50 are NMR spectra of compounds **1**–**10** (PDF)

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Notes

The authors declare no competing financial interest.

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